

3 - On using ABMs in Cognitive and Computational Social Science

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Course Outline (Modules)

- Module 1. Introduction to Multi-Agent Systems
- Module 2. Intro to Agent-Based Modeling
- **Module 3. On Using ABMs in Cognitive and Computational Social Science**
- Module 4. Exploring and Extending Agent-Based Modeling
- Module 5. Creating Agent-Based Models
- Module 7. Adaptive Agents
- Module 8. Analyzing Agent-Based Models
- Module 9. More on Analysis of ABMs
- Module 10. Example publication combining ABM, MAS, and Cognition
- Module 11. Group Projects

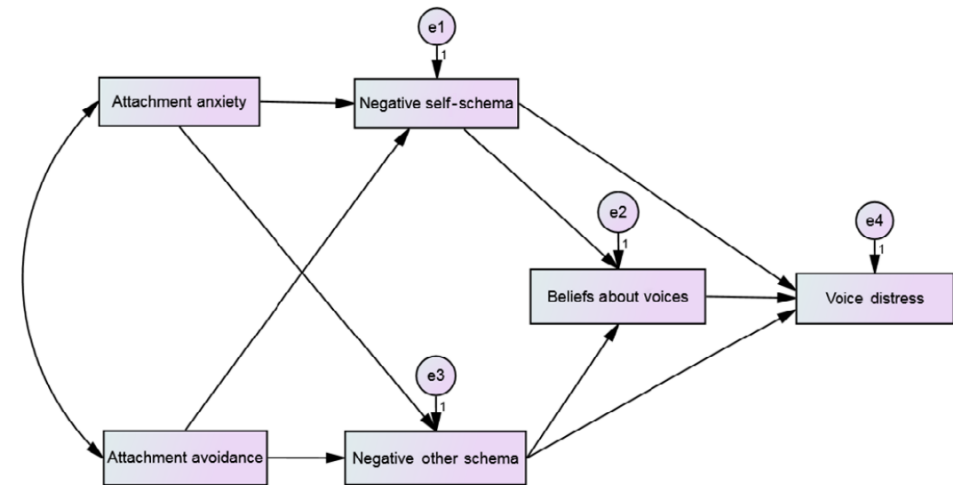
Outline

- Madsen et al. (2019)
- Conte & Paolucci (2014)

Madsen et al. (2019)

Cognitive Methods and Models

- Often developed based on cognition in isolation
 - Lab experiments, surveys, settings with high control
- Analytic models



Cole et al., 2017

Human cognition occurs in interactions

- Others influence our attention, learning, decision making, etc
- Social environments are *dynamic*
 - Interaction can create hard to predict feedback loops
- Do individual cognitive models “scale”?

The methodological challenge

- External actions may directly influence cognitive components and the behavior of cognitive entities is not trivially scaled to a population level
- Three advantage of ABMS for CogSci:
 - Explore what happens when multiple cognitive entities interact over time and space
 - Can calibrate (parameterize) and validate (test predictions) cognitive models in complex systems
 - Encourage model development

What are ABMS?

- We already covered this :)

When an ABM is a useful methodological tool?

- When cognitive function is dependent on the actions of other agents/environments
 - Contingent behaviors (where feedback influences future states)
 - E.g., Voting behavior that has micro (individual) and macro outcomes

When an ABM is a useful methodological tool?

- When there is heterogeneity within and between groups
 - Defining groups and setting properties/parameters
 - E.g., Financial market with regulators and investors
- Can be exogenous (starting with different params) vs endogenous (start with same params but diverge over time)

When an ABM is a useful methodological tool?

- When time is a factor (change over time)
 - Emergence of group dynamics
- When there is spatial distribution
 - e.g., navigation, foraging, search, interaction, etc

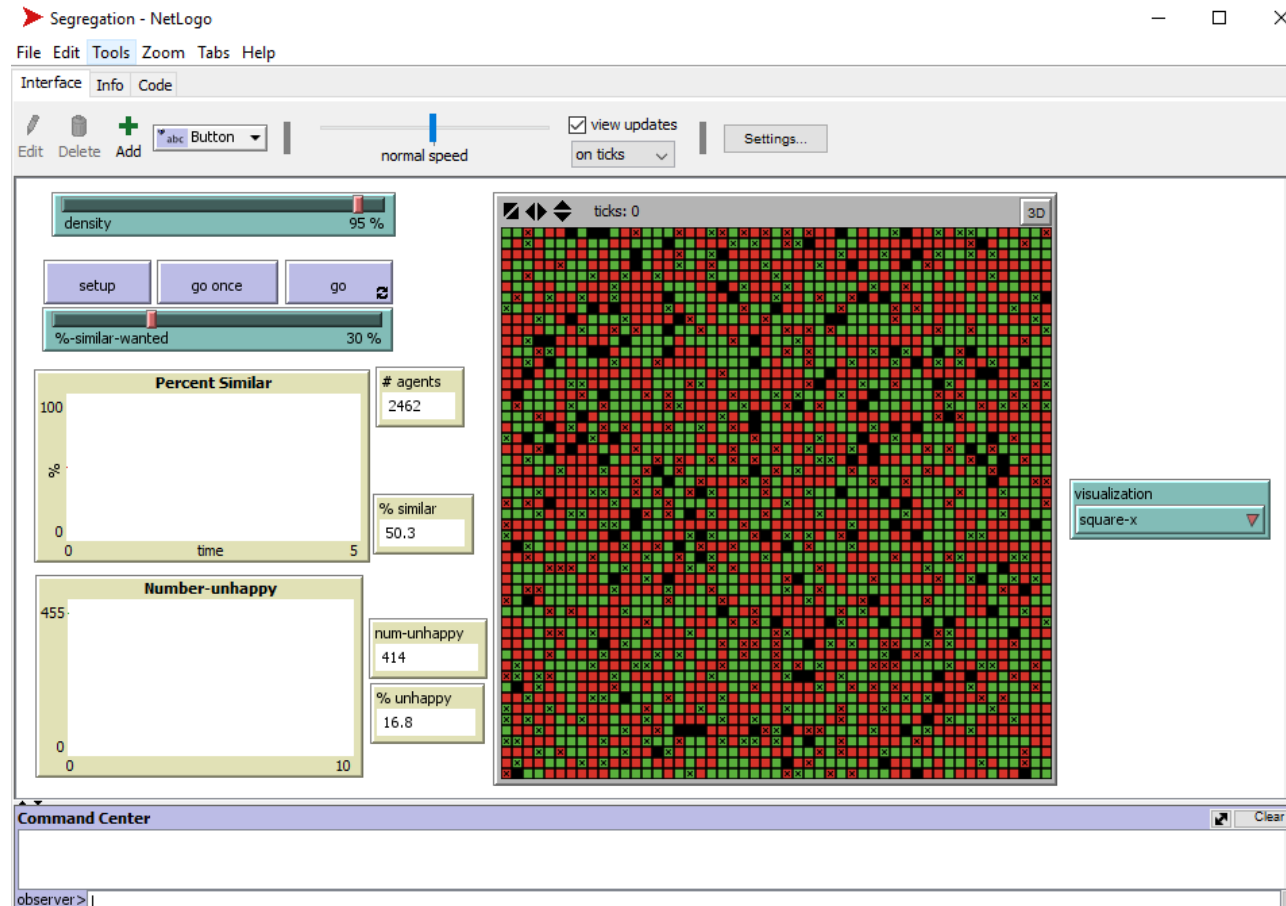
What are the uses of ABMs in CogSci?

- In CogSci, we want to understand how humans acquire knowledge, make decisions, and solve real world problems
- Calibration - Provides parameterization
 - Compared to known data to set parameters that best replicate and reproduce those data
- Validation - tests predictions
 - Parameters set theoretically, empirically, or via calibration

What are the uses of ABMs in CogSci?

- Longitudinal questions
 - Evolution of cognition, culture, social norms, etc.
- Ethical questions
 - Scenarios that are not 'ethical' to manipulate in real-life (e.g., violence, spread of misinformation, etc.)

Schelling's (1971) Segregation Model



Excercise #1

- 1) First take a look at the code of the Schelling model and try to figure out what it does
- 2) Then check the info
- 3) Is this model useful for answering CogSci questions? Why or why not? What features does it have?
- 4) Write at least one question you could answer with this model.
- 5) Run the model, and try to come up with an answer to your question.

Conte & Paolucci (2014)

Defining agents

- Autonomous systems that operate transitions between states of the world, based on mechanisms and representations incorporated into them
 - Vary in the degree of autonomy, self-interest, sociability, learning, complexity

How are mental representations incorporated (if at all)?

- Utilized to reason about the world and other agents, plan, make decisions, communicate
- Symbolic (explicitly modeled “in” an agent) vs sub-symbolic (more implicit association/relationships)

Multi-realizability

- Multi-level multi-agent systems can be implemented in different ways at lower levels and still generate the same or similar macro-level phenomena
- With varying implementations, equivalence testing is possible for structure and mechanisms of the models

Some current opportunities for ABMs

- How to scale and incorporate real-time simulation with massive amounts of data
 - Parallel and computing infrastructure
- Model equivalence
 - Have to accept multiple paths
 - Better to have strong theoretical foundations and real-world plausibility

ABM Recipe for Model Building

- **Minimality procedure** - simplest set of rules required to generate macroscopic effects
 - Seems to be consensus in model building but it has shortcomings (reduces validity) as it isn't driven by theory and can create non-plausible models

Cognitive Models

- Difficult to control the inner validity and calibration so it reflects the real-world system without also complex real-world data collection

Generative Models

- Generate theory of behavior by accounts for behavior in terms of mechanisms that are supposed to operate while producing it
- Requires
 - External (environmental and social)
 - Internal (behavioral/cognitive) mechanisms
- Aims at finding the general mechanisms yielding the wide spectrum of behaviors of relatively autonomous systems.

Computational Social Science

- Variants
 - Deductive - Explain social science phenomena with math, computer science, and logic/game theoretic models
 - Generative - ABMs as discussed
 - Complex - Combine with complex systems methods, learning, data science, machine learning, etc.

An interdisciplinary foundation for CSS

- Describe the dynamics of a given phenomena using simulated and large datasets
- Use ABM to check the internal consistency and resulting states
- Apply cross-methodological experimental methods to validate hypotheses against real-world data
- Update data-mining methods and models
- Use equation-based modeling when possible

Excercise 2

- 1) Think about the articles. What was something you didn't quite understand? What do you think was the most important point(s)?
- 2) Pair up with someone near you to discuss 1).
- 3) Share with the class

Preparing for Module 4 - Exploring and Extending ABMs

- Attend [Practical Session 2](#)
- Read: CH3, Wilensky & Rand (2015)
- Supplementary: Romanowska, I., Wren, C. D., & Crabtree, S. A. (2021). Agent-based modeling for archaeology: Simulating the complexity of societies. SFI Press. Chapter 2: Trading up to Complex Models of Economic Interaction